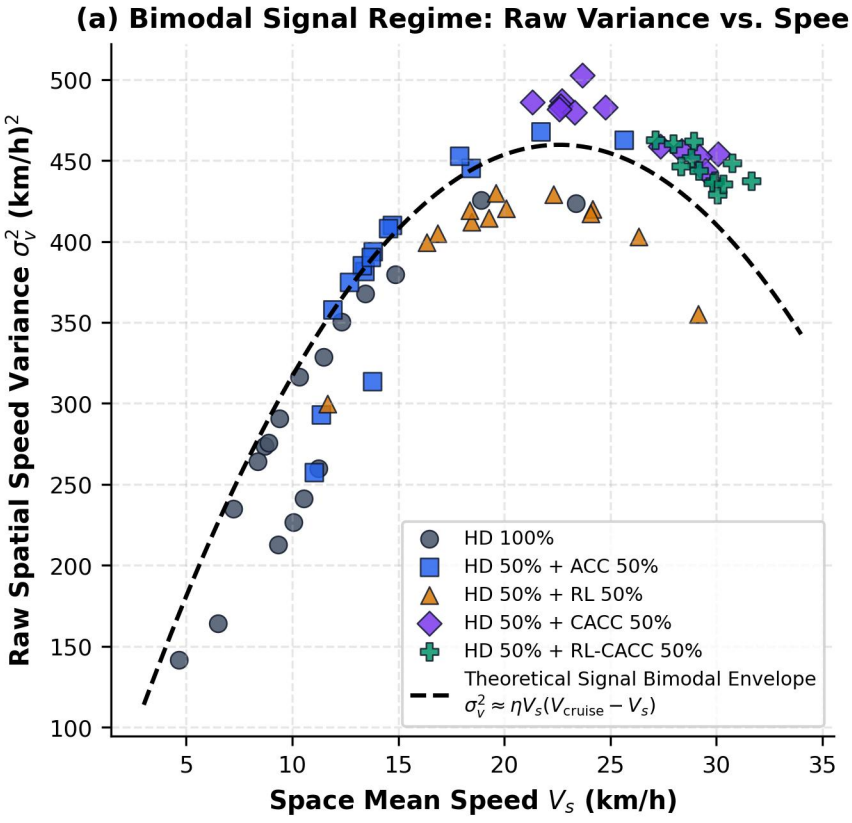
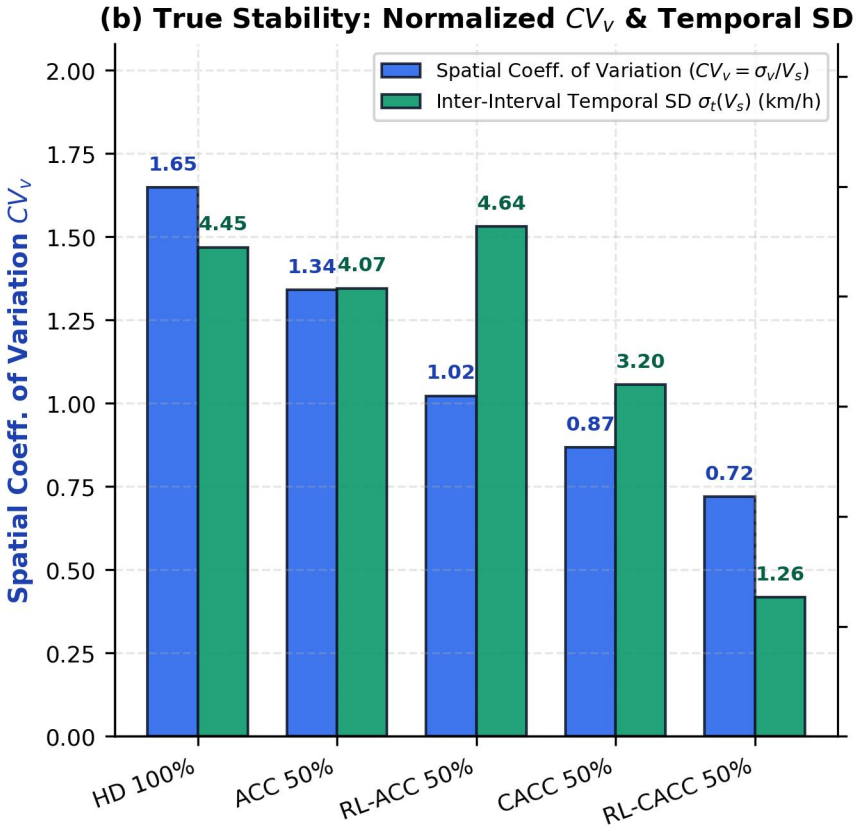
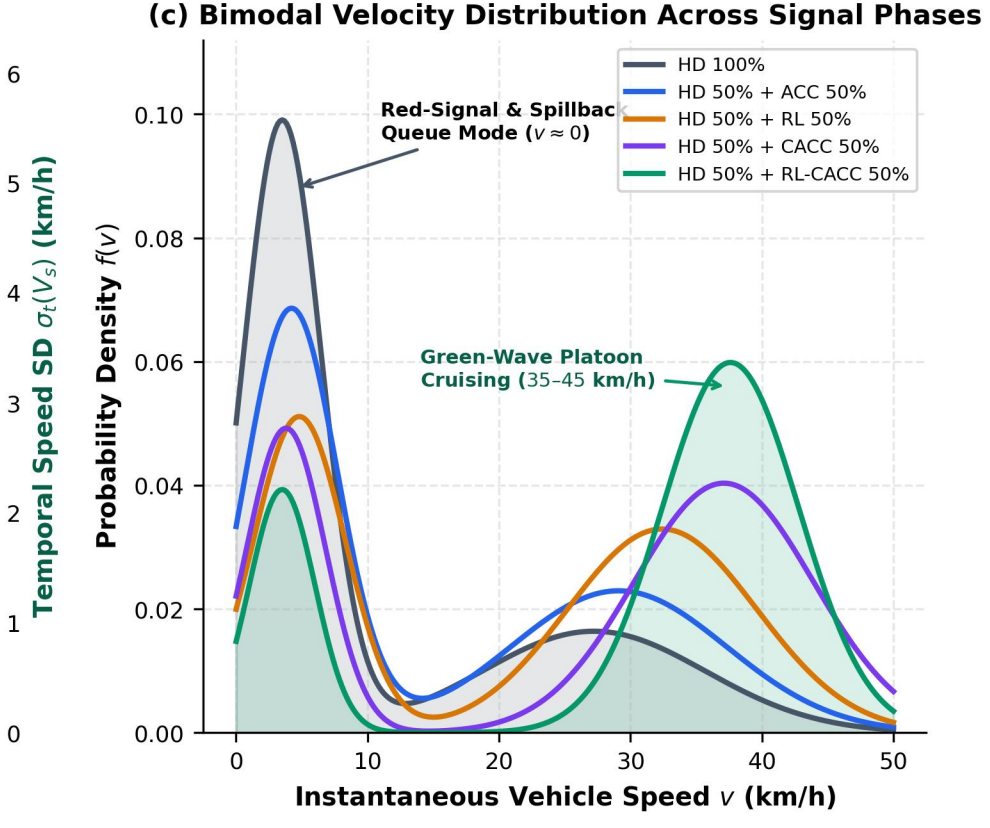

(a) Bimodal Signal Regime: Raw Variance vs. Spee d
Raw Spatial Speed Variance σ_v^2 (km/h)^2
Space Mean Speed V_s (km/h)
HD 100%
HD 50% + ACC 50%
HD 50% + RL 50%
HD 50% + CACC 50%
HD 50% + RL-CACC 50%
Theoretical Signal Bimodal Envelope
$\sigma_v^2 \approx \eta V_s(V_{cruise} - V_s)$


(b) True Stability: Normalized CV_v & Temporal SD
Spatial Coeff. of Variation ($CV_v = \sigma_v/V_s$)
Inter-Interval Temporal SD $\sigma_t(V_s)$ (km/h)
Spatial Coeff. of Variation CV_v
1.65
4.45
1.34
4.07
1.02
4.64
0.87
3.20
0.72
1.26
HD 100%
ACC 50%
RL-ACC 50%
CACC 50%
RL-CACC 50%


(c) Bimodal Velocity Distribution Across Signal Phases
Red-Signal & Spillbac
Queue Mode ($v \approx 0$)
Green-Wave Platoon
Cruising (35–45 km/h)
HD 100%
HD 50% + ACC 50%
HD 50% + RL 50%
HD 50% + CACC 50%
HD 50% + RL-CACC 50%
Probability Density $f(v)$
Temporal Speed SD $\sigma_t(V_s)$ (km/h)
Instantaneous Vehicle Speed v (km/h)